

Department: **Data Science and Artificial Intelligence**
Course Name: **Data Science Tools and Software**
Course Code: **03203N**
Professor Name: **Dr. Laila M. Fatehy**
Level: **Two**



Sheet 2
(R Programming Review)

[1] Suppose $A = \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}$

- a. Check that $A^3 = 0$ where 0 is a 3×3 matrix with every entry equal to 0 .
- b. Replace the third column of A by the sum of the second and third columns.

[2] Create the following matrix B with 15 rows:

$$B = \begin{bmatrix} 10 & -10 & 10 \\ 10 & -10 & 10 \\ \dots & \dots & \dots \\ 10 & -10 & 10 \end{bmatrix}$$

and then calculate the 3×3 matrix $B^T B$.

[3] Write R script to calculate the following:

a.
$$\sum_{i=10}^{100} (i^3 + 4i^2)$$

b.
$$2, \frac{2^2}{2}, \frac{2^3}{3}, \dots, \frac{2^{25}}{25}$$

c.
$$1 + \frac{2}{3} + \frac{4}{5} + \frac{6}{7} + \dots + \frac{38}{39}$$

[4] Describe the built-in R-data set "trees".

- a. Perform a summary of all variables in "trees ".
- b. Examine the two measures of center for the variable **Volume**
- c. Compute the mean of **Height**
- d. Which variable has the highest standard deviation
- e. What are the standard deviations of these variables

- f. Calculate the maximum and minimum value of **Girth** in one command
- g. Calculate the inter quartile range of **Volume**
- h. Construct the boxplot of the tree data, are there outliers? Explain.

[5] Describe the built-in R-data set " airquality"

- a. Create a data frame called " airquality1" and insert the airquality into it
- b. Calculate the mean and the variance of **Ozone** data
- c. Make a description statistics of the **Solar.R** data
- d. Calculate the deciles of the **Temp** data
- e. Calculate the 35th , 45th and the 67th percentiles of the **Wind** data

f. Construct a plot of **Temp** against a **Day**

g. Create a boxplot of **Month** data

h. Draw the histogram of **Ozon** data

i. Give visual summaries of all variables

Good Luck
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